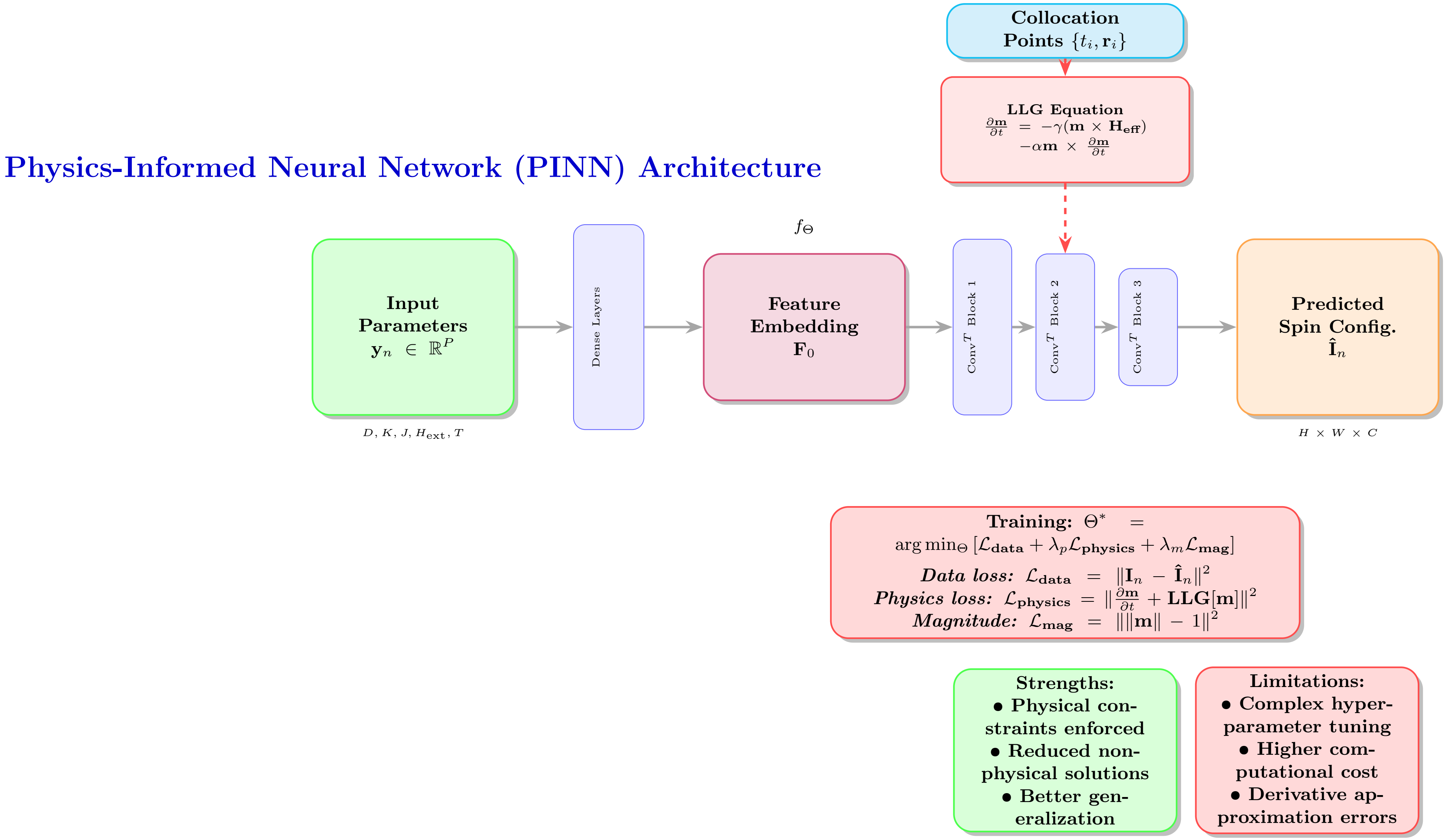

Physics-Informed Neural Network (PINN) Architecture
Collocation Points $\{t_i, \mathbf{r}_i\}$
LLG Equation
$\frac{\partial \mathbf{m}}{\partial t} = -\gamma(\mathbf{m} \times \mathbf{H}_{\text{eff}})$
$-\alpha \mathbf{m} \times \frac{\partial \mathbf{m}}{\partial t}$
$f_\Theta$
Input Parameters $\mathbf{y}_n \in \mathbb{R}^P$
$D, K, J, H_{\text{ext}}, T$
Dense Layers
Feature Embedding $\mathbf{F}_0$
Conv$^T$ Block 1
Conv$^T$ Block 2
Conv$^T$ Block 3
Predicted Spin Config. $\hat{\mathbf{I}}_n$
$H \times W \times C$
Training: $\Theta^* = \arg\min_\Theta [\mathcal{L}_{\text{data}} + \lambda_p \mathcal{L}_{\text{physics}} + \lambda_m \mathcal{L}_{\text{mag}}]$
Data loss: $\mathcal{L}_{\text{data}} = \|\mathbf{I}_n - \hat{\mathbf{I}}_n\|^2$
Physics loss: $\mathcal{L}_{\text{physics}} = \|\frac{\partial \mathbf{m}}{\partial t} + \mathbf{LLG}[\mathbf{m}]\|^2$
Magnitude: $\mathcal{L}_{\text{mag}} = \|\|\mathbf{m}\| - 1\|^2$
Strengths:
• Physical constraints enforced
• Reduced non-physical solutions
• Better generalization
Limitations:
• Complex hyperparameter tuning
• Higher computational cost
• Derivative approximation errors